

Analisis & Manajemen *Log*

Oleh:

Idris Winarno

Materi Ajar

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Pentingnya File *Log*

- File *log* seperti “kotak hitam” pada pesawat terbang.
- *Log* adalah pencatat utama dari sistem dan aktivitas jaringan.
 - Sebagai dasar pemulihan ketika terjadi perubahan layanan (secara ilegal)
 - Sebagai dasar pelacakan untuk mengetahui peyusup pada sistem kita.
- Memungkinkan untuk secara selektif memulihkan data yang rusak melalui analisa log.
- Catatan yang penting/berharga untuk keperluan bisnis lainnya.

Memahami file *log*

- File penyimpan riwayat dari aktivitas sistem.
- Menyimpan data penting tentang aktivitas sistem.
- Menyediakan petunjuk untuk investigasi
- Sebagai bukti penuntutan.

Kebijakan *log*

- Atur ulang (*reset*) file log pada interval/periode tertentu.
- Rotasi file *log*, agar data tersedia untuk durasi waktu tertentu
- Kompres dan arsip file *log* pada media penyimpanan (hdd/flashdisk).

Tipe dari *log*

- Network Devices
- Firewall
- Application Server
- System

Network Device Logs

- Router Logs
- Network Traffic
- Inbound and outbound packets
- Bandwidth Utilization
 - L3 Switch Logs
- SNMP Traps

Firewall logs

- Firewall logs menyediakan informasi penting tentang:
 - Paket yang masuk dan keluar
 - Informasi tentang server tertentu, mis. Web server
 - Paket yang di blok/drop
 - Alarm untuk SysAdmin
 - Penyadapan sistem
 - dll

Application Server Logs

- Web Server logs
 - Error Logs
 - Access Logs
- Mail Server logs
 - Connection Status
 - SMTP queues
 - Protocol Status (IMAP, POP3, SMTP)
- FTP Server Logs
 - Current logins
 - Commands executed
 - File uploaded and downloaded
- Database Server Logs
 - User activity
 - Objects accessed
 - Creation of new tables, databases, etc..

System Logs

- Security Logs
 - Authentication logs
 - Object Database access logs
 - Files and Folder auditing
- Application Logs
 - Access Logs
 - Error Logs
- System Logs
 - Kernel dumps
 - Memory Dump logs
 - Driver initialization failed

Manajemen *Log* (Laporan)

- *Wikipedia:*

“Log management (LM) comprises an approach to dealing with large volumes of computer-generated log messages (also known as audit records, audit trails, event-logs, etc.).”

- Tujuan dari manajemen log adalah tentang perhatian terhadap keamanan, operasi sistem dan jaringan, serta kepatuhan terhadap regulasi/peraturan.

Manajemen *Log*

- Log Management meliputi:
 - Log collection
 - Centralized log aggregation
 - Long-term log storage and retention
 - Log rotation
 - Log analysis (in real-time and in bulk after storage)
 - Log search and reporting.

Masalah pada *log*

- Kegagalan untuk memahami pentingnya *log* menyebabkan manajemen *log* yang buruk.
- Tidak adanya *review* oleh SysAdmin.
- *Log* tidak berjalan karena ukuran disk yang terbatas.
- Ketika file *log* yang berhubungan dengan penyerangan terhapus, maka akan sulit untuk mengetahui apa yang telah termodifikasi dan mustahil untuk melacak penyerang.

Tantangan utama Log

- Volume

Data log bisa mencapai ratusan GB per hari dan tantangannya adalah mengkoleksi/menyimpan data tersebut.

- Normalization

Data log dihasilkan dengan berbagai macam format. Proses normalisasi digunakan untuk proses analisa lebih lanjut.

- Velocity

Kecepatan dalam menghasilkan log mengakibatkan kesulitan dalam koleksi dan agregasi.

- Veracity

Aktifitas yang dilaporkan di-log mungkin tidak akurat, terutama permasalahan dari sistem-sistem yang menjalankan sistem deteksi seperti IDS.

Fungsi Log

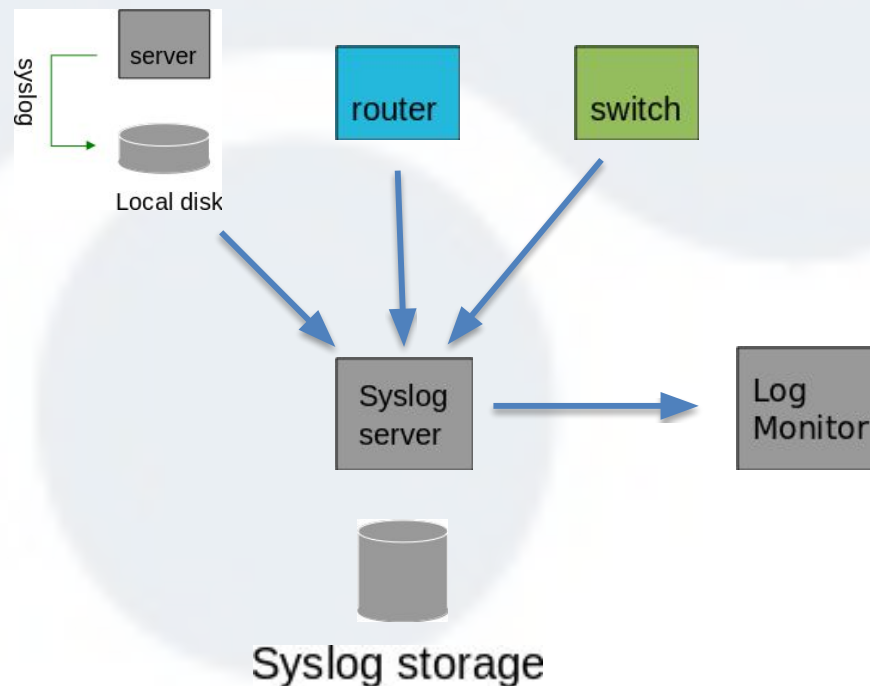
- Dapat melaporkan adanya *bug* (yang sebenarnya)
- Membantu menganalisa, reproduksi dan menyelesaikan *bug*
- Membantu ujicoba fitur-fitur baru pada fase *development*

Terminologi

- Logging
 - Segera/langsung dihapus
 - Representasi dari pemrosesan data/input dari pengguna
- Auditing:
 - Tidak segera/langsung dihapus
 - Data disimpan dan dilindungi dengan skema keamanan tertentu
 - Selalu terhubung ke beberapa fungsi *end-user*

Arsitektur

- Log Generation
 - Komputer/Perangkat yang menghasilkan Log
- Log Analysis and Storage
 - Komputer/Server yang menerima data/copy log
- Log Monitoring
 - Komputer yang berfungsi untuk memonitor/*review* data log menjadi sebuah laporan.



Amankan file *log*

- Setel hak akses yang layak untuk file *log*
- Gunakan mesin server untuk *log* yang terpisah (Central Syslogger)
- Salin file *log* secara teratur.
- Gunakan media yang hanya bisa ditulis sekali.
- Enkripsi-lah file *log*.

Syslog (Centralized Logging)

- Dikembangkan sejak 1980-an
- Adalah sebuah aplikasi dan protocol standar untuk mentransfer log.
- Format Syslog:
 - Tipe pesan (*facility*): misalnya pesan kernel, mail, auth, dll
 - *Severity*: nilai/tingkat dari pesan (0: emergency, 7:debug)

```
Mar  1 06:25:43 server1 sshd[23170]: Accepted publickey for server2 from 172.30.128.115 port 21011 ssh2

Mar  1 07:16:42 server1 sshd[9326]: Accepted password for murugiah from 10.20.30.108 port 1070 ssh2

Mar  1 07:16:53 server1 sshd[22938]: reverse mapping checking getaddrinfo for ip10.165.nist.gov failed - POSSIBLE BREAKIN ATTEMPT!

Mar  1 07:26:28 server1 sshd[22572]: Accepted publickey for server2 from 172.30.128.115 port 30606 ssh2

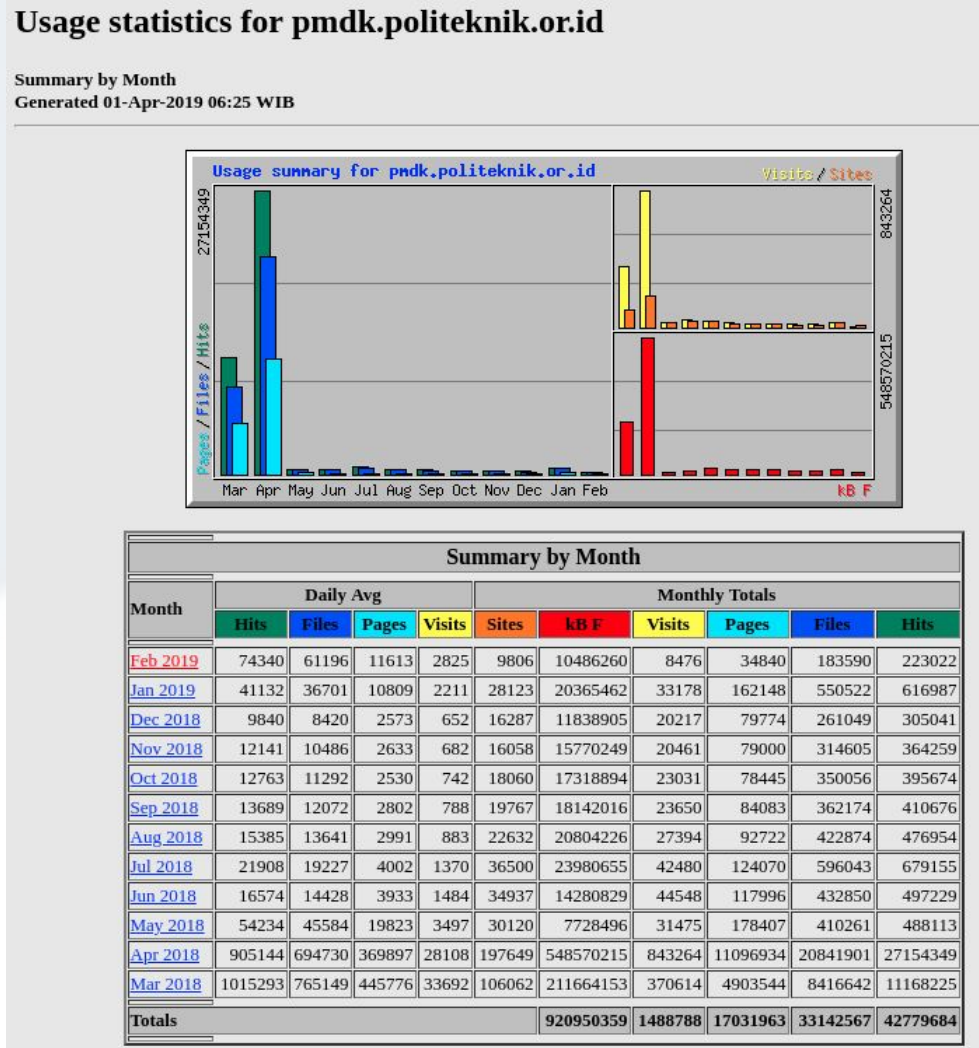
Mar  1 07:28:33 server1 su: BAD SU kkent to root on /dev/tty2

Mar  1 07:28:41 server1 su: kkent to root on /dev/tty2
```

Value	Severity	Keyword	Deprecated keywords	Description
0	Emergency	emerg	panic ^[7]	System is unusable
1	Alert	alert		Action must be taken immediately
2	Critical	crit		Critical conditions
3	Error	err	error ^[7]	Error conditions
4	Warning	warning	warn ^[7]	Warning conditions
5	Notice	notice		Normal but significant conditions
6	Informational	info		Informational messages
7	Debug	debug		Debug-level messages

Contoh1: Aplikasi analisa log web server

- Sering disebut juga **web log analyzer**
- **Webalizer**



Contoh2: Aplikasi analisa proxy

- SARG: Squid Analysis Report Generator



SARG Squid Analysis Report Generator

Squid User Access Reports
Period: 2019Mar31-2019Mar31
Sort: BYTES, reverse
Topuser

[Topsites](#)
[Sites & Users](#)
[Downloads](#)
[Denied](#)

NUM		USERID	CONNECT	BYTES	%BYTES	IN-CACHE-OUT	ELAPSED TIME	MILISEC	%TIME
1		10.252.39.109	2.71K	54.78G	17.77%	0.01% 3.92%	31:37:42	113,862,689	0.32%
2		10.252.39.130	2.16K	25.11G	8.15%	0.00% 8.55%	82:59:52	298,792,522	0.85%
3		10.252.130.128	4.47K	19.05G	6.18%	0.02% 11.27%	139:19:26	501,566,745	1.42%
4		10.252.133.42	3.04K	7.44G	2.42%	0.11% 28.84%	46:09:14	166,154,015	0.47%
5		10.252.133.27	14.81K	6.10G	1.98%	0.01% 35.17%	356:39:22	1,283,962,207	3.64%
6		10.252.131.7	5.67K	6.07G	1.97%	0.02% 35.37%	166:44:19	600,259,159	1.70%
7		10.252.132.133	751	5.88G	1.91%	0.00% 36.50%	21:16:55	76,615,145	0.22%
8		10.252.62.163	18.12K	5.38G	1.75%	0.01% 39.91%	241:44:35	870,275,565	2.47%
9		10.252.111.138	13.08K	4.50G	1.46%	0.24% 47.67%	250:44:06	902,646,644	2.56%
10		10.252.128.51	7.69K	4.34G	1.41%	0.01% 49.40%	70:37:12	254,232,806	0.72%
11		10.252.2.99	9.05K	4.19G	1.36%	0.15% 51.14%	137:10:15	493,815,808	1.40%
12		10.252.129.179	18.80K	4.02G	1.31%	4.11% 53.30%	71:06:07	255,967,419	0.73%

Contoh3: LogAnalyzer

The screenshot displays the LogAnalyzer web application interface. At the top, there's a navigation bar with links for Search, Show Events, Statistics, Reports, Help, Search in Knowledge Base, and Maximize View. A search bar is prominently featured with a 'Search (filter):' label and a 'Search' button. Below the search bar, a table titled 'Recent syslog messages' lists various system events. The table has columns for Date, Facility, Severity, Host, Syslogtag, ProcessID, Messagetype, and Message. The messages include kernel logs, network interface status changes, and system boot logs. The interface also includes a 'Select Language' dropdown set to 'English' and a 'Donate' button. The bottom of the screen shows a Windows taskbar with the Start button and several application icons.

Date	Facility	Severity	Host	Syslogtag	ProcessID	Messagetype	Message
Today 15:35:18			ubuntu	kernel --- [249.090453] en...		Syslog	I/O error, dev fd0, sector 0
Today 15:35:18			ubuntu	kernel --- [249.070454] en...		Syslog	I/O error, dev fd0, sector 0
Today 15:31:24			ubuntu	kernel --- [19.903292] nb...		Syslog	registered device at major 43
Today 15:31:24			ubuntu	kernel --- [19.635562] IP...		Syslog	ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
Today 15:31:24			ubuntu	kernel --- [19.619243] e1...		Syslog	eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: None
Today 15:31:24			ubuntu	kernel --- [19.601775] IP...		Syslog	ADDRCONF(NETDEV_UP): eth0: link is not ready
Today 15:31:24			ubuntu	kernel --- [19.601035] IP...		Syslog	ADDRCONF(NETDEV_UP): eth0: link is not ready
Today 15:31:23			ubuntu	kernel --- [18.658911] ty...		Syslog	11): apparmor="STATUS" operation="profile_load" profile="unconfined" name="/usr/ ...
Today 15:31:21			ubuntu	kernel --- [17.378861] ty...		Syslog	10): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="/u ...
Today 15:31:21			ubuntu	kernel --- [17.378636] ty...		Syslog	9): apparmor="STATUS" operation="profile_load" profile="unconfined" name="/usr/s ...
Today 15:31:21			ubuntu	kernel --- [17.378631] ty...		Syslog	8): apparmor="STATUS" operation="profile_load" profile="unconfined" name="/usr/l ...
Today 15:31:21			ubuntu	kernel --- [17.228085] Bl...		Syslog	BNEP socket layer initialized
Today 15:31:21			ubuntu	kernel --- [17.228076] Bl...		Syslog	BNEP filters: protocol multicast
Today 15:31:21			ubuntu	kernel --- [17.228073] Bl...		Syslog	BNEP (Ethernet Emulation) ver 1.3
Today 15:31:21			ubuntu	kernel --- [17.199920] Bl...		Syslog	RFCOMM ver 1.11
Today 15:31:21			ubuntu	kernel --- [17.199915] Bl...		Syslog	RFCOMM socket layer initialized
Today 15:31:21			ubuntu	kernel --- [17.199892] Bl...		Syslog	RFCOMM TTY layer initialized
Today 15:31:21			ubuntu	kernel --- [17.174451] Bl...		Syslog	SCO socket layer initialized
Today 15:31:21			ubuntu	kernel --- [17.174449] Bl...		Syslog	L2CAP socket layer initialized
Today 15:31:21			ubuntu	kernel --- [17.174448] Bl...		Syslog	HCI socket layer initialized
Today 15:31:21			ubuntu	kernel --- [17.174440] Bl...		Syslog	HCI device and connection manager initialized
Today 15:31:21			ubuntu	kernel --- [17.174439] NE...		Syslog	Registered protocol family 31
Today 15:31:21			ubuntu	kernel --- [17.174408] Bl...		Syslog	Core ver 2.17
Today 15:31:21			ubuntu	kernel --- [17.102037] [d...		Syslog	00:0f:0 on minor 0
Today 15:31:20			ubuntu	rsyslogd-2212 - - imtcp		Syslog	module loaded, but no listeners defined - no input will be gathered [v8.11.0.ma ...
Today 15:31:20			ubuntu	kernel --- [16.520994] Co...		Syslog	switching to colour frame buffer device 100x37

TERIMA KASIH